BRINGING REGO AND RELATIONSHIPS TOGETHER

Topaz (https://www.topaz.sh) is an open-source authorization service that blends two powerful ideas: the flexibility of Open Policy Agent (OPA) and the graph-based relationship modeling of Google Zanzibar. Instead of choosing between attribute-based rules and relationship-based access, Topaz allows you to use both.

At its core, Topaz evaluates policies written in Rego, OPA's policy language. This allows you to express complex, context-rich rules such as time-based access, IP restrictions, or approval workflows. In addition to policy evaluation, Topaz maintains a directory of relationships between objects, principals, and groups, like a Zanzibar-style ReBAC system. This enables queries like "Can Carol edit documents owned by people she manages?" to be both efficient and straightforward.

Topaz is designed for cloud-native environments: it can run as a microservice, a sidecar, or in a serverless workflow, and supports real-time, low-latency decision-making. Policies can be stored in version control, embedded in OCI images, and deployed through CI/CD pipelines. The project also includes developer-friendly tools such as a web console, relationship visualizer, and detailed decision logs.

For organizations aiming to combine ABAC and ReBAC without adopting two separate systems, Topaz provides an integrated model.

8.4 AWS IAM

Launched in 2011, AWS Identity and Access Management (IAM, https://aws.amazon.com/iam/) is one of the most widely used authorization systems in the world. Although it's solely connected to AWS, millions of developers use it daily to manage access to their AWS services. If you've launched an EC2 instance, uploaded data to S3, or called a Lambda function, you've used IAM.